

# Stakeholder Engagement and Ground Research

## Overview

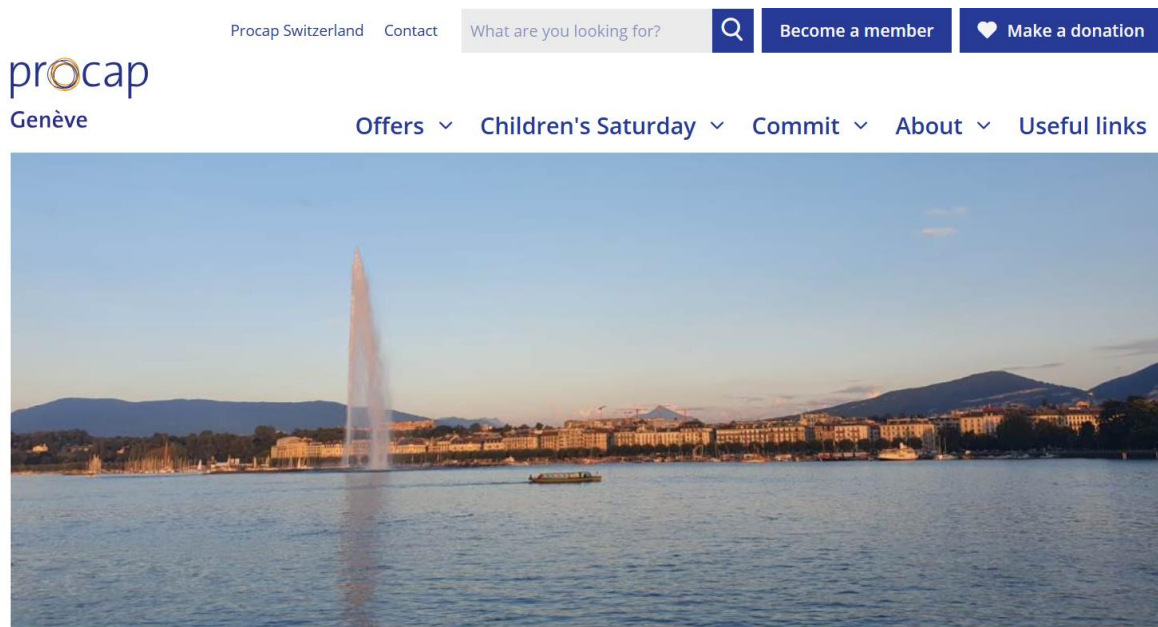
AiBility was developed with the objective of improving accessibility and inclusivity within public transportation systems. From the outset, we recognized that accessibility requirements should not be defined solely through technical assumptions. Instead, the project adopted a user-centred approach by engaging directly with organizations and individuals who experience accessibility challenges in their daily mobility.

To better understand these challenges, the team conducted stakeholder engagement activities with local accessibility advocacy and support organizations in Geneva. These discussions focused on mobility barriers, accessibility information needs, communication preferences, and expectations for digital transportation services. The insights gathered during this process directly influenced the crowdsourcing and AI components of the platform.

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## Partner Organizations

### Procap Genève



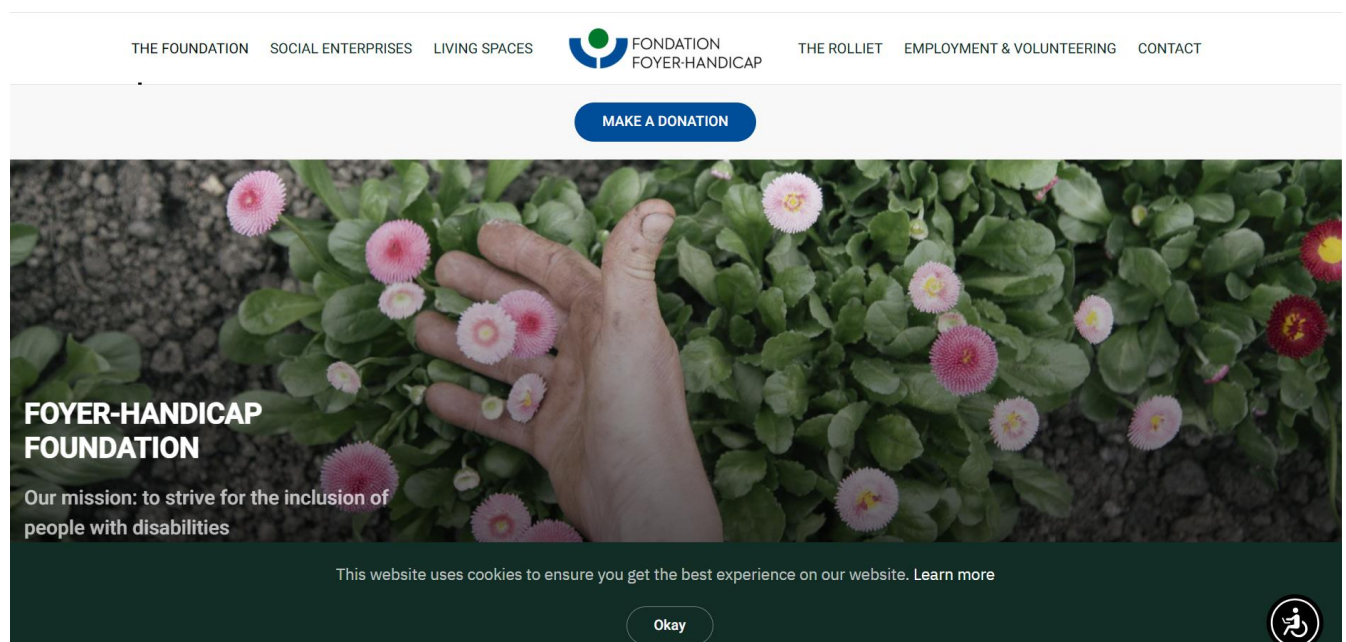
<https://www.procap-geneve.ch/>

Procap Genève is the Geneva branch of Procap Switzerland, one of the country's leading organizations supporting and advocating for people with disabilities. Through legal support, accessibility advocacy, community engagement, and social inclusion initiatives, Procap works to promote equal participation and independent living.

Discussions with members of Procap Genève provided valuable perspectives on the everyday mobility challenges encountered when navigating public transportation systems. These conversations highlighted the importance of reliable accessibility information, independence during travel, and inclusive service design.

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## Fondation Foyer-Handicap



<https://foyer-handicap.ch/>

Fondation Foyer-Handicap is a Geneva-based organization dedicated to supporting the autonomy, inclusion, employment, mobility, and quality of life of people with disabilities. The foundation provides adapted living environments, transportation services, employment opportunities, and community support programs designed to encourage independent participation in society.

Our visit to Fondation Foyer-Handicap enabled direct engagement with residents and staff representing a broad range of accessibility needs. These discussions provided practical insights into transportation barriers, digital accessibility, navigation challenges, and the role of technology in supporting independent mobility.

## Stakeholder Representation

The engagement activities included participants representing diverse accessibility needs, including:

- Wheelchair users and individuals with mobility impairments.
- People who are blind or have visual impairments.
- Individuals with hearing impairments.
- People with neurodiverse conditions, including autism spectrum conditions.
- Individuals with upper-limb differences or reduced dexterity.
- Support staff and caregivers working within disability support organizations.

These discussions reinforced that accessibility is not a single design problem. Different users experience transportation systems differently and therefore benefit from different forms of support, information, and accommodation.

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## Key Findings

### Accessibility Conditions Change Over Time

Participants emphasized that accessibility cannot be treated as a fixed property of a transportation network. Infrastructure maintenance, temporary construction, equipment failures, weather conditions, and operational disruptions can significantly alter the accessibility of a stop or route.

This observation later motivated both the CrowdSense accessibility evaluation component and the exploration of accessibility dynamics as an emerging research direction within the project.

### Information Gaps Create Uncertainty

Many participants reported that uncertainty itself often becomes a barrier to travel. Inaccurate or outdated accessibility information can make journey planning difficult and reduce confidence in using public transportation independently.

This finding highlighted the need for continuously updated accessibility information generated through community participation and crowdsourcing.

### Mobility Requires More Than Transportation

For many users, accessibility challenges begin before boarding a vehicle. Locating stops, identifying vehicles, navigating unfamiliar environments, and responding to unexpected disruptions are all important components of the travel experience.

This feedback influenced the development of voice-assisted interaction and navigation support features within AiBility.

## Simplicity Improves Accessibility

Participants consistently emphasized the importance of intuitive interfaces and clear communication. Digital services that require extensive navigation or complex interactions may unintentionally create additional barriers for users with different accessibility needs.

This insight guided the design of the simplified booking workflow and accessibility-focused user experience.

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## Impact on System Design

The stakeholder engagement process directly informed several major design decisions within AiBility.

| Stakeholder Insight                                  | Design Response                          |
|------------------------------------------------------|------------------------------------------|
| Accessibility conditions change dynamically          | CrowdSense accessibility evaluations     |
| Accessibility information is often outdated          | Community-driven accessibility reporting |
| Users require different accessibility accommodations | Personalized accessibility profiles      |
| Navigation extends beyond booking a trip             | Voice-guided navigation assistance       |
| Simple interactions improve usability                | Voice-first booking experience           |
| Accessibility challenges vary across user groups     | Inclusive and adaptive interface design  |

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## Key Learnings

The most significant lesson from the stakeholder engagement process was that accessibility should be viewed as a dynamic and multi-dimensional challenge rather than a static infrastructure characteristic.

The discussions demonstrated that meaningful accessibility solutions require a combination of reliable information, user participation, inclusive design, and adaptive technology. They also highlighted the value of crowdsourcing as a mechanism for capturing accessibility experiences that may change over time and differ across user groups.

These insights shaped the broader vision of AiBility: a platform that combines crowdsourcing, artificial intelligence, and user-centred design to support more inclusive public transportation.

## Acknowledgements

The AiBility team would like to sincerely thank Procap Genève and Fondation Foyer-Handicap for their time, support, and willingness to engage with our project.

We are especially grateful to the residents, members, staff, caregivers, and participants who shared their experiences and perspectives during our stakeholder engagement activities. Their feedback provided invaluable insight into the challenges associated with accessible mobility and helped ensure that the project remained grounded in real user needs rather than technical assumptions.

Many of the platform's key features including accessibility profiles, voice-based interaction, CrowdSense accessibility evaluations, and the Accessibility Dynamics research initiative were directly influenced by the experiences and recommendations shared during these discussions.

Their contributions played a fundamental role in shaping the development of AiBility, and we greatly appreciate the trust and support they extended to our team.